

Sets and Indices

T, t	Set and index for 15-minute time intervals $t \in \{1, \dots, 35040\}$
B, b	Set and index for 4-hour ancillary service blocks $b \in \{1, \dots, 2190\}$
J, j	Set and index for cycle degradation cost segments $j \in \{1, \dots, J\}$
I, i	Set and index for calendar degradation cost breakpoints $i \in \{1, \dots, I\}$

0.0.1 Parameters

Symbol	Description	Unit
$P_{DA}(t)$	Day-ahead electricity price	EUR/MWh
$P_{FCR}(b)$	FCR capacity price	EUR/MW
$P_{aFRR}^{\text{pos}}(b)$	Positive aFRR capacity price	EUR/MW
$P_{aFRR}^{\text{neg}}(b)$	Negative aFRR capacity price	EUR/MW
$P_{aFRR,E}^{\text{pos}}(t)$	(New) Positive aFRR energy price	EUR/MWh
$P_{aFRR,E}^{\text{neg}}(t)$	(New) Negative aFRR energy price	EUR/MWh
E_{nom}	Nominal energy capacity	kWh
$P_{\text{max}}^{\text{config}}$	Max charge/discharge power	kW
$\eta_{\text{ch}}, \eta_{\text{dis}}$	Charging/discharging efficiencies	-
$SOC_{\text{min}}, SOC_{\text{max}}$	Min/max SOC limits (fraction)	-
E_j^{seg}	(New) Max energy capacity of segment j	kWh
c_j^{cost}	(New) Marginal cyclic degradation cost for segment j	EUR/kWh
SOC_i^{point}	(New) SOC breakpoint i for calendar cost	kWh
$Cost_i^{\text{point}}$	(New) Calendar degradation cost at breakpoint i	EUR/hr
α	(New) Degradation price (meta-parameter)	-

0.0.2 Decision Variables

Symbol	Description	Type
$p_{\text{ch}}(t), p_{\text{dis}}(t)$	DA charge/discharge power	Cont. ≥ 0
$p_{aFRR,E}^{\text{pos}}(t)$	(New) aFRR-E positive (discharge) power	Cont. ≥ 0
$p_{aFRR,E}^{\text{neg}}(t)$	(New) aFRR-E negative (charge) power	Cont. ≥ 0
$c_{fcr}(b)$	FCR capacity bid	Cont. ≥ 0
$c_{aFRR}^{\text{pos}}(b)$	Positive aFRR capacity bid	Cont. ≥ 0
$c_{aFRR}^{\text{neg}}(b)$	Negative aFRR capacity bid	Cont. ≥ 0
$p_{\text{ch}}^{\text{total}}(t), p_{\text{dis}}^{\text{total}}(t)$	(New) Total charge/discharge power	Cont. ≥ 0
$p_j^{\text{ch}}(t), p_j^{\text{dis}}(t)$	(New) Charge/discharge power for segment j	Cont. ≥ 0
$e_{\text{soc},j}(t)$	(New) Energy stored in segment j	Cont. ≥ 0
$e_{\text{soc}}(t)$	(New) Total energy stored in BESS	Cont. ≥ 0
$c_{\text{cost}}^{\text{cal}}(t)$	(New) Calendar cost at time t	Cont. ≥ 0
$\lambda_{t,i}$	(New) SOS2 variable for calendar cost	Cont. ≥ 0
$y_{\text{ch}}(t), y_{\text{dis}}(t)$	DA bid binaries	Binary
$y_{fcr}(b), y_{aFRR}^{\text{pos}}(b), \dots$	Ancillary service bid binaries	Binary
$y_{aFRR,E}^{\text{pos}}(t), \dots$	(New) aFRR-E bid binaries	Binary
$y_{\text{ch}}^{\text{total}}(t), y_{\text{dis}}^{\text{total}}(t)$	(New) Total operation binaries	Binary

1 Model (i): Base Model + aFRR Energy Market

1.1 Objective Function

The objective is to maximize the total profit from market participation.

$$\max Z = \mathbb{P}^{DA} + \mathbb{P}^{ANCI} + \mathbb{P}^{aFRR_E} \quad (1)$$

Where:

$$\mathbb{P}^{DA} = \sum_{t \in T} \left(\frac{P_{DA}(t)}{1000} p_{\text{dis}}(t) - \frac{P_{DA}(t)}{1000} p_{\text{ch}}(t) \right) \Delta t \quad (2)$$

$$\mathbb{P}^{ANCI} = \sum_{b \in B} \left(P_{FCR}(b) c_{fcr}(b) + P_{aFRR}^{\text{pos}}(b) c_{aFRR}^{\text{pos}}(b) + P_{aFRR}^{\text{neg}}(b) c_{aFRR}^{\text{neg}}(b) \right) \quad (3)$$

$$\mathbb{P}^{aFRR_E} = \sum_{t \in T} \left(\frac{P_{aFRR,E}^{\text{pos}}(t)}{1000} p_{aFRR,E}^{\text{pos}}(t) - \frac{P_{aFRR,E}^{\text{neg}}(t)}{1000} p_{aFRR,E}^{\text{neg}}(t) \right) \Delta t \quad (4)$$

1.2 Constraints

1.2.1 Standard SOC Dynamics

$$e_{\text{soc}}(t) = e_{\text{soc}}(t-1) + \left(p_{\text{ch}}^{\text{total}}(t) \eta_{\text{ch}} - \frac{p_{\text{dis}}^{\text{total}}(t)}{\eta_{\text{dis}}} \right) \Delta t \quad \forall t \quad (5)$$

$$p_{\text{ch}}^{\text{total}}(t) = p_{\text{ch}}(t) + p_{aFRR,E}^{\text{neg}}(t) \quad \forall t \quad (6)$$

$$p_{\text{dis}}^{\text{total}}(t) = p_{\text{dis}}(t) + p_{aFRR,E}^{\text{pos}}(t) \quad \forall t \quad (7)$$

1.2.2 SOC Limits

$$SOC_{\min} E_{\text{nom}} \leq e_{\text{soc}}(t) \leq SOC_{\max} E_{\text{nom}} \quad \forall t \quad (8)$$

1.2.3 Simultaneous Operation Prevention

$$p_{\text{ch}}^{\text{total}}(t) \leq y_{\text{ch}}^{\text{total}}(t) \cdot P_{\max}^{\text{config}} \quad \forall t \quad (9)$$

$$p_{\text{dis}}^{\text{total}}(t) \leq y_{\text{dis}}^{\text{total}}(t) \cdot P_{\max}^{\text{config}} \quad \forall t \quad (10)$$

$$y_{\text{ch}}^{\text{total}}(t) + y_{\text{dis}}^{\text{total}}(t) \leq 1 \quad \forall t \quad (11)$$

1.2.4 Market Co-optimization Power Limits

$$p_{\text{dis}}^{\text{total}}(t) + 1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{pos}}(b) \leq P_{\max}^{\text{config}} \quad \forall b, t \in b \quad (12)$$

$$p_{\text{ch}}^{\text{total}}(t) + 1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{neg}}(b) \leq P_{\max}^{\text{config}} \quad \forall b, t \in b \quad (13)$$

1.2.5 Ancillary Service Energy Reserve

$$\frac{(1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{pos}}(b)) \tau}{\eta_{\text{dis}}} \leq e_{\text{soc}}(t) - SOC_{\min} E_{\text{nom}} \quad \forall b, t \in b \quad (14)$$

$$(1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{neg}}(b)) \tau \eta_{\text{ch}} \leq SOC_{\max} E_{\text{nom}} - e_{\text{soc}}(t) \quad \forall b, t \in b \quad (15)$$

(Where τ is the reserve duration, e.g., 0.25h)

1.2.6 Ancillary Capacity Market Exclusivity

$$y_{fcr}(b) + y_{aFRR}^{\text{pos}}(b) + y_{aFRR}^{\text{neg}}(b) \leq 1 \quad \forall b \quad (16)$$

1.2.7 Cross-Market Mutual Exclusivity

$$y_{\text{ch}}^{\text{total}}(t) + y_{fcr}(b) + y_{aFRR}^{\text{pos}}(b) \leq 1 \quad \forall b, t \in b \quad (17)$$

$$y_{\text{dis}}^{\text{total}}(t) + y_{fcr}(b) + y_{aFRR}^{\text{neg}}(b) \leq 1 \quad \forall b, t \in b \quad (18)$$

1.2.8 Minimum Bids & Binary Logic

$$y_{\text{ch}}(t) \text{MinBid}_{da} \cdot 1000 \leq p_{\text{ch}}(t) \leq y_{\text{ch}}(t) P_{\text{max}}^{\text{config}} \quad \forall t \quad (19)$$

$$y_{\text{dis}}(t) \text{MinBid}_{da} \cdot 1000 \leq p_{\text{dis}}(t) \leq y_{\text{dis}}(t) P_{\text{max}}^{\text{config}} \quad \forall t \quad (20)$$

$$y_{aFRR,E}^{\text{pos}}(t) \text{MinBid}_{afrr-e} \cdot 1000 \leq p_{aFRR,E}^{\text{pos}}(t) \leq y_{aFRR,E}^{\text{pos}}(t) P_{\text{max}}^{\text{config}} \quad \forall t \quad (21)$$

$$y_{aFRR,E}^{\text{neg}}(t) \text{MinBid}_{afrr-e} \cdot 1000 \leq p_{aFRR,E}^{\text{neg}}(t) \leq y_{aFRR,E}^{\text{neg}}(t) P_{\text{max}}^{\text{config}} \quad \forall t \quad (22)$$

$$y_{\text{ch}}^{\text{total}}(t) \geq y_{\text{ch}}(t); \quad y_{\text{ch}}^{\text{total}}(t) \geq y_{aFRR,E}^{\text{neg}}(t) \quad \forall t \quad (23)$$

$$y_{\text{dis}}^{\text{total}}(t) \geq y_{\text{dis}}(t); \quad y_{\text{dis}}^{\text{total}}(t) \geq y_{aFRR,E}^{\text{pos}}(t) \quad \forall t \quad (24)$$

(Similar min/max bid constraints for c_{fcr} , c_{aFRR}^{pos} , c_{aFRR}^{neg} remain from Phase I)

2 Model (ii): Model (i) + Cyclic Aging Cost

2.1 Objective Function

The objective is to maximize the total profit, defined as market revenues minus cyclic degradation costs.

$$\max Z = \mathbb{P}^{DA} + \mathbb{P}^{ANCI} + \mathbb{P}^{aFRR-E} - \alpha \cdot C^{\text{cyc}} \quad (25)$$

Where:

$$\mathbb{P}^{DA} = \sum_{t \in T} \left(\frac{P_{DA}(t)}{1000} p_{\text{dis}}(t) - \frac{P_{DA}(t)}{1000} p_{\text{ch}}(t) \right) \Delta t \quad (26)$$

$$\mathbb{P}^{ANCI} = \sum_{b \in B} \left(P_{fcr}(b) c_{fcr}(b) + P_{aFRR}^{\text{pos}}(b) c_{aFRR}^{\text{pos}}(b) + P_{aFRR}^{\text{neg}}(b) c_{aFRR}^{\text{neg}}(b) \right) \quad (27)$$

$$\mathbb{P}^{aFRR-E} = \sum_{t \in T} \left(\frac{P_{aFRR,E}^{\text{pos}}(t)}{1000} p_{aFRR,E}^{\text{pos}}(t) - \frac{P_{aFRR,E}^{\text{neg}}(t)}{1000} p_{aFRR,E}^{\text{neg}}(t) \right) \Delta t \quad (28)$$

$$C^{\text{cyc}} = \sum_{t \in T} \sum_{j \in J} \left(c_j^{\text{cost}} \cdot \frac{p_j^{\text{dis}}(t)}{\eta_{\text{dis}}} \cdot \Delta t \right) \quad (29)$$

2.2 Constraints

2.2.1 Degradation-Aware SOC Dynamics

$$e_{\text{soc},j}(t) = e_{\text{soc},j}(t-1) + \left(p_j^{\text{ch}}(t) \eta_{\text{ch}} - \frac{p_j^{\text{dis}}(t)}{\eta_{\text{dis}}} \right) \Delta t \quad \forall t, \forall j \quad (30)$$

$$e_{\text{soc}}(t) = \sum_{j \in J} e_{\text{soc},j}(t) \quad \forall t \quad (31)$$

$$p_{\text{ch}}^{\text{total}}(t) = \sum_{j \in J} p_j^{\text{ch}}(t) \quad \forall t \quad (32)$$

$$p_{\text{dis}}^{\text{total}}(t) = \sum_{j \in J} p_j^{\text{dis}}(t) \quad \forall t \quad (33)$$

$$p_{\text{ch}}^{\text{total}}(t) = p_{\text{ch}}(t) + p_{aFRR,E}^{\text{neg}}(t) \quad \forall t \quad (34)$$

$$p_{\text{dis}}^{\text{total}}(t) = p_{\text{dis}}(t) + p_{aFRR,E}^{\text{pos}}(t) \quad \forall t \quad (35)$$

2.2.2 SOC & Segment Limits

$$SOC_{\min} E_{\text{nom}} \leq e_{\text{soc}}(t) \leq SOC_{\max} E_{\text{nom}} \quad \forall t \quad (36)$$

$$0 \leq e_{\text{soc},j}(t) \leq E_j^{\text{seg}} \quad \forall t, \forall j \quad (37)$$

2.2.3 Simultaneous Operation Prevention

$$p_{\text{ch}}^{\text{total}}(t) \leq y_{\text{ch}}^{\text{total}}(t) \cdot P_{\max}^{\text{config}} \quad \forall t \quad (38)$$

$$p_{\text{dis}}^{\text{total}}(t) \leq y_{\text{dis}}^{\text{total}}(t) \cdot P_{\max}^{\text{config}} \quad \forall t \quad (39)$$

$$y_{\text{ch}}^{\text{total}}(t) + y_{\text{dis}}^{\text{total}}(t) \leq 1 \quad \forall t \quad (40)$$

2.2.4 Market Co-optimization Power Limits

$$p_{\text{dis}}^{\text{total}}(t) + 1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{pos}}(b) \leq P_{\max}^{\text{config}} \quad \forall b, t \in b \quad (41)$$

$$p_{\text{ch}}^{\text{total}}(t) + 1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{neg}}(b) \leq P_{\max}^{\text{config}} \quad \forall b, t \in b \quad (42)$$

2.2.5 Ancillary Service Energy Reserve

$$\frac{(1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{pos}}(b)) \tau}{\eta_{\text{dis}}} \leq e_{\text{soc}}(t) - SOC_{\min} E_{\text{nom}} \quad \forall b, t \in b \quad (43)$$

$$(1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{neg}}(b)) \tau \eta_{\text{ch}} \leq SOC_{\max} E_{\text{nom}} - e_{\text{soc}}(t) \quad \forall b, t \in b \quad (44)$$

(Where τ is the reserve duration, e.g., 0.25h)

2.2.6 Ancillary Capacity Market Exclusivity

$$y_{fcr}(b) + y_{aFRR}^{\text{pos}}(b) + y_{aFRR}^{\text{neg}}(b) \leq 1 \quad \forall b \quad (45)$$

2.2.7 Cross-Market Mutual Exclusivity

$$y_{\text{ch}}^{\text{total}}(t) + y_{fcr}(b) + y_{aFRR}^{\text{pos}}(b) \leq 1 \quad \forall b, t \in b \quad (46)$$

$$y_{\text{dis}}^{\text{total}}(t) + y_{fcr}(b) + y_{aFRR}^{\text{neg}}(b) \leq 1 \quad \forall b, t \in b \quad (47)$$

2.2.8 Minimum Bids & Binary Logic

$$y_{\text{ch}}(t) \text{MinBid}_{da} \cdot 1000 \leq p_{\text{ch}}(t) \leq y_{\text{ch}}(t) P_{\text{max}}^{\text{config}} \quad \forall t \quad (48)$$

$$y_{\text{dis}}(t) \text{MinBid}_{da} \cdot 1000 \leq p_{\text{dis}}(t) \leq y_{\text{dis}}(t) P_{\text{max}}^{\text{config}} \quad \forall t \quad (49)$$

$$y_{aFRR,E}^{\text{pos}}(t) \text{MinBid}_{aFRR_e} \cdot 1000 \leq p_{aFRR,E}^{\text{pos}}(t) \leq y_{aFRR,E}^{\text{pos}}(t) P_{\text{max}}^{\text{config}} \quad \forall t \quad (50)$$

$$y_{aFRR,E}^{\text{neg}}(t) \text{MinBid}_{aFRR_e} \cdot 1000 \leq p_{aFRR,E}^{\text{neg}}(t) \leq y_{aFRR,E}^{\text{neg}}(t) P_{\text{max}}^{\text{config}} \quad \forall t \quad (51)$$

$$y_{\text{ch}}^{\text{total}}(t) \geq y_{\text{ch}}(t); \quad y_{\text{ch}}^{\text{total}}(t) \geq y_{aFRR,E}^{\text{neg}}(t) \quad \forall t \quad (52)$$

$$y_{\text{dis}}^{\text{total}}(t) \geq y_{\text{dis}}(t); \quad y_{\text{dis}}^{\text{total}}(t) \geq y_{aFRR,E}^{\text{pos}}(t) \quad \forall t \quad (53)$$

(Similar min/max bid constraints for c_{fcr} , c_{aFRR}^{pos} , c_{aFRR}^{neg} remain from Phase I)

3 Model (iii): Model (ii) + Calendar Aging Cost

3.1 Objective Function

The objective is to maximize the total profit, defined as market revenues minus degradation costs.

$$\max Z = \mathbb{P}^{DA} + \mathbb{P}^{ANCI} + \mathbb{P}^{aFRR_E} - \alpha \cdot (C^{\text{cyc}} + C^{\text{cal}}) \quad (54)$$

Where:

$$\mathbb{P}^{DA} = \sum_{t \in T} \left(\frac{P_{DA}(t)}{1000} p_{\text{dis}}(t) - \frac{P_{DA}(t)}{1000} p_{\text{ch}}(t) \right) \Delta t \quad (55)$$

$$\mathbb{P}^{ANCI} = \sum_{b \in B} \left(P_{FCR}(b) c_{fcr}(b) + P_{aFRR}^{\text{pos}}(b) c_{aFRR}^{\text{pos}}(b) + P_{aFRR}^{\text{neg}}(b) c_{aFRR}^{\text{neg}}(b) \right) \quad (56)$$

$$\mathbb{P}^{aFRR_E} = \sum_{t \in T} \left(\frac{P_{aFRR,E}^{\text{pos}}(t)}{1000} p_{aFRR,E}^{\text{pos}}(t) - \frac{P_{aFRR,E}^{\text{neg}}(t)}{1000} p_{aFRR,E}^{\text{neg}}(t) \right) \Delta t \quad (57)$$

$$C^{\text{cyc}} = \sum_{t \in T} \sum_{j \in J} \left(c_j^{\text{cost}} \cdot \frac{p_j^{\text{dis}}(t)}{\eta_{\text{dis}}} \cdot \Delta t \right) \quad (58)$$

$$C^{\text{cal}} = \sum_{t \in T} c_{\text{cost}}^{\text{cal}}(t) \cdot \Delta t \quad (59)$$

3.2 Constraints

3.2.1 Degradation-Aware SOC Dynamics (Replaces Cst-1)

$$e_{\text{soc},j}(t) = e_{\text{soc},j}(t-1) + \left(p_j^{\text{ch}}(t) \eta_{\text{ch}} - \frac{p_j^{\text{dis}}(t)}{\eta_{\text{dis}}} \right) \Delta t \quad \forall t, \forall j \quad (60)$$

$$e_{\text{soc}}(t) = \sum_{j \in J} e_{\text{soc},j}(t) \quad \forall t \quad (61)$$

$$p_{\text{ch}}^{\text{total}}(t) = \sum_{j \in J} p_j^{\text{ch}}(t) \quad \forall t \quad (62)$$

$$p_{\text{dis}}^{\text{total}}(t) = \sum_{j \in J} p_j^{\text{dis}}(t) \quad \forall t \quad (63)$$

$$p_{\text{ch}}^{\text{total}}(t) = p_{\text{ch}}(t) + p_{aFRR,E}^{\text{neg}}(t) \quad \forall t \quad (64)$$

$$p_{\text{dis}}^{\text{total}}(t) = p_{\text{dis}}(t) + p_{aFRR,E}^{\text{pos}}(t) \quad \forall t \quad (65)$$

3.2.2 SOC & Segment Limits (Replaces Cst-2)

$$SOC_{\min} E_{\text{nom}} \leq e_{\text{soc}}(t) \leq SOC_{\max} E_{\text{nom}} \quad \forall t \quad (66)$$

$$0 \leq e_{\text{soc},j}(t) \leq E_j^{\text{seg}} \quad \forall t, \forall j \quad (67)$$

3.2.3 Simultaneous Operation Prevention (Replaces Cst-3)

$$p_{\text{ch}}^{\text{total}}(t) \leq y_{\text{ch}}^{\text{total}}(t) \cdot P_{\max}^{\text{config}} \quad \forall t \quad (68)$$

$$p_{\text{dis}}^{\text{total}}(t) \leq y_{\text{dis}}^{\text{total}}(t) \cdot P_{\max}^{\text{config}} \quad \forall t \quad (69)$$

$$y_{\text{ch}}^{\text{total}}(t) + y_{\text{dis}}^{\text{total}}(t) \leq 1 \quad \forall t \quad (70)$$

3.2.4 Market Co-optimization Power Limits (Replaces Cst-4)

$$p_{\text{dis}}^{\text{total}}(t) + 1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{pos}}(b) \leq P_{\max}^{\text{config}} \quad \forall b, t \in b \quad (71)$$

$$p_{\text{ch}}^{\text{total}}(t) + 1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{neg}}(b) \leq P_{\max}^{\text{config}} \quad \forall b, t \in b \quad (72)$$

3.2.5 Ancillary Service Energy Reserve (Keeps Cst-6 Logic)

$$\frac{(1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{pos}}(b)) \tau}{\eta_{\text{dis}}} \leq e_{\text{soc}}(t) - SOC_{\min} E_{\text{nom}} \quad \forall b, t \in b \quad (73)$$

$$(1000 c_{fcr}(b) + 1000 c_{aFRR}^{\text{neg}}(b)) \tau \eta_{\text{ch}} \leq SOC_{\max} E_{\text{nom}} - e_{\text{soc}}(t) \quad \forall b, t \in b \quad (74)$$

(Where τ is the reserve duration, e.g., 0.25h)

3.2.6 Ancillary Capacity Market Exclusivity (Keeps Cst-7)

$$y_{fcr}(b) + y_{aFRR}^{\text{pos}}(b) + y_{aFRR}^{\text{neg}}(b) \leq 1 \quad \forall b \quad (75)$$

3.2.7 Cross-Market Mutual Exclusivity (Replaces Cst-8)

$$y_{\text{ch}}^{\text{total}}(t) + y_{fcr}(b) + y_{aFRR}^{\text{pos}}(b) \leq 1 \quad \forall b, t \in b \quad (76)$$

$$y_{\text{dis}}^{\text{total}}(t) + y_{fcr}(b) + y_{aFRR}^{\text{neg}}(b) \leq 1 \quad \forall b, t \in b \quad (77)$$

3.2.8 Minimum Bids & Binary Logic (Updates Cst-9)

$$y_{\text{ch}}(t) \text{MinBid}_{da} \cdot 1000 \leq p_{\text{ch}}(t) \leq y_{\text{ch}}(t) P_{\max}^{\text{config}} \quad \forall t \quad (78)$$

$$y_{\text{dis}}(t) \text{MinBid}_{da} \cdot 1000 \leq p_{\text{dis}}(t) \leq y_{\text{dis}}(t) P_{\max}^{\text{config}} \quad \forall t \quad (79)$$

$$y_{aFRR,E}^{\text{pos}}(t) \text{MinBid}_{afrr_e} \cdot 1000 \leq p_{aFRR,E}^{\text{pos}}(t) \leq y_{aFRR,E}^{\text{pos}}(t) P_{\max}^{\text{config}} \quad \forall t \quad (80)$$

$$y_{aFRR,E}^{\text{neg}}(t) \text{MinBid}_{afrr_e} \cdot 1000 \leq p_{aFRR,E}^{\text{neg}}(t) \leq y_{aFRR,E}^{\text{neg}}(t) P_{\max}^{\text{config}} \quad \forall t \quad (81)$$

$$y_{\text{ch}}^{\text{total}}(t) \geq y_{\text{ch}}(t); \quad y_{\text{ch}}^{\text{total}}(t) \geq y_{aFRR,E}^{\text{neg}}(t) \quad \forall t \quad (82)$$

$$y_{\text{dis}}^{\text{total}}(t) \geq y_{\text{dis}}(t); \quad y_{\text{dis}}^{\text{total}}(t) \geq y_{aFRR,E}^{\text{pos}}(t) \quad \forall t \quad (83)$$

(Similar min/max bid constraints for c_{fcr} , c_{aFRR}^{pos} , c_{aFRR}^{neg} remain from Phase I)

3.2.9 Calendar Aging PWL Constraints (New)

$$e_{\text{soc}}(t) = \sum_{i \in I} \lambda_{t,i} \cdot SOC_i^{\text{point}} \quad \forall t \quad (84)$$

$$c_{\text{cost}}^{\text{cal}}(t) = \sum_{i \in I} \lambda_{t,i} \cdot Cost_i^{\text{point}} \quad \forall t \quad (85)$$

$$\sum_{i \in I} \lambda_{t,i} = 1 \quad \forall t \quad (86)$$

$$\{\lambda_{t,i}\}_{i \in I} \text{ are SOS2 variables} \quad \forall t \quad (87)$$